Milestone 3: Econometric Models Grading Rubric

QM 2023 Capstone Project

Total Points: 50 **Grading Mode:** Team grade (all members receive same score unless peer evaluation indicates disparity)

Component Breakdown

1. Model Specification (15 points)

Excellent (13-15 points):

- ☑ **Model A (Fixed Effects):** Correctly estimated with `entity_effects=True` and `time_effects=True`
- ☑ **Model B:** Chosen specification (DiD/ARIMA/ML) is appropriate for research question
- ☑ **Variable selection:** Predictors are economically sensible (e.g., 2-month lag for FEDFUNDS based on M2 EDA)
- ☑ **Clustered standard errors:** Used correctly (`cov_type='clustered', cluster_entity=True`)
- ☑ **Code structure:** Clear sections, comments explain logic, reproducible

Satisfactory (9-12 points):

- ☑ Both models estimated, but minor specification issues (e.g., forgot time FE, or used wrong lag)
- ☑ Variable selection reasonable but not optimal
- ☑ Standard errors computed but not clustered (heteroskedasticity concern)
- ☑ Code works but lacks clear comments

Needs Improvement (5-8 points):

- ⚠ One model estimated correctly, the other has errors
- ⚠ Variable selection is arbitrary (no justification from M2 or theory)
- ⚠ No robust standard errors (ignored heteroskedasticity)
- ⚠ Code is difficult to follow (poor structure)

Unsatisfactory (0-4 points):

- ❌ Neither model estimated correctly (wrong syntax, output errors)
- ❌ Models are nonsensical (e.g., time series regression on panel data)
- ❌ Code does not run

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2. Diagnostics & Robustness (12 points)

Diagnostics (6 points):

Excellent (6 points):

- ☑ **Heteroskedasticity test:** Breusch-Pagan or visual inspection (residuals vs. fitted plot)

- ☑ **Multicollinearity (VIF):** Calculated for all predictors; high VIF addressed (variable dropped or combined)
- ☑ **Residual plots:** Q-Q plot or histogram shows approximate normality (or fat tails acknowledged)
- ☑ **All diagnostics interpreted:** Not just reported, but explained (e.g., "VIF > 10 for SIZE, so we dropped it")

Satisfactory (4-5 points):

- ☑ 2 of 3 diagnostics run (e.g., VIF + residual plot, but no heteroskedasticity test)
- ☑ Diagnostics reported but interpretation is weak

Needs Improvement (2-3 points):

- ⚠ Only 1 diagnostic run
- ⚠ Diagnostics run but not interpreted

Unsatisfactory (0-1 points):

- ✕ No diagnostics run

Robustness Checks (6 points):

Excellent (6 points):

- ☑ **At least 3 robustness checks conducted:**
  - Alternative lag structures (test lags 1, 2, 3)
  - Exclusion of outlier periods (e.g., COVID crash)
  - Sector subsamples (estimate model separately for sensitive vs. resilient sectors)
  - Placebo test (for DiD: run regression on pre-treatment period)
- ☑ **Results compared:** "Baseline model: β = -0.025. Excluding COVID: β = -0.023. Results are robust."
- ☑ **Implications discussed:** "Alternative lags yield similar coefficients, confirming lag 2 is not arbitrary."

Satisfactory (4-5 points):

- ☑ 2 robustness checks conducted
- ☑ Results compared but implications not fully discussed

Needs Improvement (2-3 points):

- ⚠ Only 1 robustness check
- ⚠ Checks conducted but results not compared

Unsatisfactory (0-1 points):

- ✕ No robustness checks

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3. Interpretation & Economic Reasoning (18 points)

Coefficient Interpretation (10 points):

Excellent (9-10 points):

- ☑ **Coefficients interpreted in economic terms:** "A 1 pp increase in FEDFUNDS reduces REIT returns by 2.5 pp"
- ☑ **Magnitude assessed:** "During 2022-2023 rate hike (5.25 pp), model predicts 13% decline"
- ☑ **Sign and significance discussed:** "Negative coefficient ($\beta < 0$) is statistically significant ($p = 0.012$) and economically sensible (leverage channel)"
- ☑ **Factor premiums explained:** "MOM coefficient (0.042) indicates trend-following profits"
- ☑ **Non-significant results acknowledged:** "SIZE is not significant ($p = 0.465$) after controlling for factors and FE"

Satisfactory (6-8 points):

- ☑ Coefficients interpreted but not always in economic terms ("$\beta = -0.025$" without explaining what this means)
- ☑ Magnitude mentioned but not assessed for economic importance
- ☑ Sign and significance noted but not explained WHY

Needs Improvement (3-5 points):

- ⚠ Minimal interpretation ("Coefficient is negative and significant")
- ⚠ No discussion of magnitude or economic mechanism

Unsatisfactory (0-2 points):

- ❌ No interpretation provided
- ❌ Interpretation is incorrect (e.g., "1% increase" vs. "1 pp increase")

Economic Mechanisms & Theory (8 points):

Excellent (7-8 points):

- ☑ **Economic mechanisms explained:** Leverage channel (higher rates → higher financing costs), Discount rate channel (higher rates → lower NPV of cash flows)
- ☑ **Consistent with theory:** Results align with financial economics (e.g., Modigliani-Miller, DCF valuation)
- ☑ **Heterogeneity discussed:** "Sector differences reflect variation in leverage, lease structure, and demand elasticity"
- ☑ **Alternative explanations considered:** "Could omitted variable (investor sentiment) confound results?"

Satisfactory (5-6 points):

- ☑ Some economic reasoning provided but not comprehensive
- ☑ Theory mentioned but not deeply integrated

Needs Improvement (2-4 points):

- ⚠ Minimal economic reasoning (just statistical descriptions)
- ⚠ No connection to theory

Unsatisfactory (0-1 points):

- ❌ No economic reasoning (purely mechanical interpretation)

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4. Presentation & Documentation (5 points)

Regression Tables (3 points):

Excellent (3 points):

- ☑ **Publication-ready tables:** Formatted with clear columns (Variable, Coefficient, SE, t-stat, p-value)
- ☑ **Comparison table:** Side-by-side for Model A and Model B
- ☑ **Notes row:** Indicates FE, time FE, clustered SEs, N, R^2
- ☑ **Significance stars:** *** $p<0.01$, ** $p<0.05$, * $p<0.10$

Satisfactory (2 points):

- ☑ Tables present but formatting could be improved
- ☑ Comparison table missing or incomplete

Needs Improvement (1 point):

- ⚠ Tables are raw Python output (not formatted)

Unsatisfactory (0 points):

- ❌ No tables or tables are unreadable

Code & Memo Quality (2 points):

Excellent (2 points):

- ☑ **Code:** Clean, modular, well-commented, reproducible
- ☑ **M3_interpretation.md:** Complete, professional, addresses all required sections (findings, diagnostics, robustness, caveats)

Satisfactory (1 point):

- ☑ Code works but lacks comments
- ☑ Interpretation memo present but incomplete

Needs Improvement (0 points):

- ❌ Code is messy or hard to follow
- ❌ Memo missing or minimal

AI Audit Appendix Requirement (Pass/Fail, Zero Credit if Missing)

Required: `AI_AUDIT_APPENDIX.md` documenting all AI tool use.

Pass (Credit for M3):

- ☑ All AI tools used are listed

- ☑ Specific prompts for econometric decisions (e.g., "How do I interpret a Fixed Effects coefficient?")
- ☑ AI output shown (code snippets, interpretations)
- ☑ Verification step documented (ran code, checked diagnostics)
- ☑ Critique provided (corrected AI misinterpretations)

Fail (Zero Credit for M3):

- ❌ No AI Audit Appendix submitted
- ❌ Appendix is generic with no details

Note: Even if models are perfect, **missing AI Audit = 0/50 points** per course policy.

Bonus Points (Optional, up to +5 points)

Exceptional work beyond requirements:

- **+2 points:** Three-way Fixed Effects (REIT + Time + Sector-Time interactions)
- **+2 points:** Modern DiD methods (Callaway & Sant'Anna for staggered adoption)
- **+1 point:** Heterogeneous treatment effects (DiD by sector or size subsample)
- **+1 point:** Monte Carlo simulation or bootstrapped standard errors

Total Possible: 55 points (50 base + 5 bonus)

Example Grade Calculations

Team A: Excellent Work (47/50 + 2 bonus = 49/50)

- Model Specification: 15/15 (both models perfect, code clean)
- Diagnostics & Robustness: 11/12 (3 diagnostics + 3 robustness checks; 1 check could be deeper)
- Interpretation: 17/18 (strong economic reasoning, minor gap in alternative explanations)
- Presentation: 4/5 (tables excellent, memo strong but minor typo)
- Bonus: +2 (three-way FE implemented)
- **Grade:** 49/50 (A+)

Team B: Satisfactory Work (38/50)

- Model Specification: 11/15 (both models run, but clustered SEs forgotten initially)
- Diagnostics & Robustness: 8/12 (2 diagnostics, 2 robustness checks)
- Interpretation: 15/18 (coefficients interpreted, but economic mechanisms could be deeper)
- Presentation: 4/5 (tables good, memo complete)
- **Grade:** 38/50 (B)

Team C: Needs Improvement (26/50)

- Model Specification: 8/15 (FE correct, DiD has specification error)
- Diagnostics & Robustness: 5/12 (1 diagnostic, 1 robustness check)
- Interpretation: 10/18 (basic interpretation, no economic reasoning)

- Presentation: 3/5 (tables are raw output, memo incomplete)
- **Grade:** 26/50 (D+)

Team D: Unsatisfactory (0/50)

- Missing AI Audit Appendix
- **Grade:** 0/50 (automatic zero per course policy)

Rubric Summary Table

Component	Excellent	Satisfactory	Needs Improvement	Unsatisfactory
Model Specification (15)	13-15	9-12	5-8	0-4
Diagnostics (6)	6	4-5	2-3	0-1
Robustness (6)	6	4-5	2-3	0-1
Interpretation (10)	9-10	6-8	3-5	0-2
Economic Reasoning (8)	7-8	5-6	2-4	0-1
Presentation (5)	4-5	3	1-2	0
Total (50)	45-50	32-44	17-31	0-16
AI Audit	Pass (required)	Pass (required)	Pass (required)	Fail = 0/50
Bonus (optional)	+0 to +5	-	-	-

Grading Timeline

- **Submission Deadline:** Friday, Week 12 (Apr 17) by 11:59 PM
- **Auto-Grading:** GitHub Actions checks code execution
- **Instructor Grading:** Completed within 1 week (by Friday, Week 13)
- **Feedback Posted:** Blackboard rubric + GitHub comments

How to Use This Rubric

For Students (Self-Assessment Before Submission)

Checklist:

- ☐ Both Model A (FE) and Model B (DiD/ARIMA/ML) estimated correctly
- ☐ Clustered standard errors used in Model A
- ☐ At least 3 diagnostics run (heteroskedasticity, VIF, residual plots)
- ☐ At least 3 robustness checks conducted (alternative lags, exclusions, subsamples)
- ☐ All coefficients interpreted in economic terms (not just statistical)
- ☐ Economic mechanisms explained (leverage, discount rates, etc.)
- ☐ Regression tables are publication-ready (formatted, not raw output)

- ☐ M3_interpretation.md complete with findings, diagnostics, robustness, caveats
- ☐ AI_AUDIT_APPENDIX.md submitted and detailed
- ☐ Code runs from scratch without errors

Estimated Score Calculator:

> *"Model A perfect (15/15), Model B minor issue (12/15 = 13). Diagnostics solid (6/6), robustness good (5/6). Interpretation strong (9/10 + 7/8 = 16/18). Tables formatted (4/5). Total: 15 + 6 + 5 + 16 + 4 = 46/50. We should add one more robustness check to get full points."*

For Instructors (Grading Guide)

1. **Auto-check:** Did code execute without errors?
2. **Model verification:** Run code manually; check that FE and DiD/ARIMA/ML are correctly specified
3. **Diagnostics review:** Verify tests ran and were interpreted (not just copy-pasted)
4. **Interpretation quality:** Do students explain WHY coefficients are economically meaningful?
5. **AI Audit check:** Is appendix present and detailed?
6. **Feedback:** Provide specific, actionable comments

Example Feedback:

> *"Your FE model is excellent (clustered SEs, correct specification). However, your DiD model has a parallel trends violation (pre-trend divergence visible in plot). Consider adding a placebo test to assess this concern. Also, your interpretation of the SIZE coefficient needs economic reasoning—WHY is it not significant after controlling for factors? (-2 points Model Specification, -2 points Interpretation)"*

Frequently Asked Questions

Q: We only did 2 robustness checks instead of 3. Will we lose points?

A: Robustness Checks section = 6 points. With 2 checks (instead of 3), you'll receive 4-5/6 points (Satisfactory). **Lost: 1-2 points.**

Q: Our DiD model failed the placebo test (found significant effect in pre-period). Is that okay?

A: No. A failed placebo test suggests parallel trends are violated, which undermines the causal interpretation of your DiD estimate. **Options:**

1. Acknowledge this limitation in M3_interpretation.md and discuss why DiD may not be appropriate for your data.
2. Choose a different Model B (ARIMA or ML comparison instead of DiD).
3. Restrict DiD to a subsample where parallel trends are more plausible.

Grading impact: If you acknowledge and address the violation, you'll receive partial credit. If you ignore it, you'll lose points for Economic Reasoning.

Q: Can we use AI to help interpret our regression coefficients?

A: Yes, but you must **verify** that AI's interpretation is correct. Example:

- **AI output:** "A 1% increase in FEDFUNDS reduces returns by 2.5%."
- **Your verification:** "This phrasing is ambiguous. Correct: 'A 1 percentage point increase in FEDFUNDS reduces returns by 2.5 percentage points.'"

Document this in AI_AUDIT_APPENDIX.md. If you don't verify and submit incorrect interpretations, you'll lose points.

Q: We have heteroskedasticity but forgot to use clustered SEs. Can we resubmit?

A: No. Milestone grades are final. **However**, if you catch this before the deadline, you can fix it and resubmit (no penalty if within deadline window).

Grading impact: Without clustered SEs, you'll lose points in Model Specification (down from 15/15 to ~11/15) and Diagnostics (down from 6/6 to ~4/6). **Total lost: ~6 points.**

Q: Our R^2 is 0.35. Will we lose points?

A: No. R^2 measures fit, not coefficient validity. What matters:

- Are coefficients statistically significant? (t-stats, p-values)
- Are they economically sensible? (signs, magnitudes)
- Are they robust? (alternative specs yield similar results)

Low R^2 is common in financial data. **Explain this in M3_interpretation.md:** "While $R^2 = 0.35$ indicates that 35% of return variation is explained, our coefficient estimates are precise and robust. Financial returns are inherently noisy due to unobserved factors (sentiment, liquidity), so moderate R^2 is expected."

Grading Philosophy

This rubric emphasizes:

1. **Methodological rigor:** Did you test assumptions and apply correct fixes?
2. **Economic intuition:** Do you understand WHY results make sense?
3. **Honest self-assessment:** Did you acknowledge limitations?
4. **Reproducibility:** Can someone else run your code and get identical results?

We are not grading:

- Statistical sophistication (advanced models beyond course scope)
- Perfect R^2 (unrealistic for financial data)
- Length of code (concise is fine)

We are grading:

- Did you apply course concepts correctly?
- Can you defend your methodological choices?
- Do you understand the economic implications of your results?

Rubric prepared for QM 2023: Statistics II, Spring 2026, University of Tulsa Contact: Dr. Cayman Seagraves (cayman-seagraves@utulsa.edu)